

Series

A **sequence** is an ordered, countably infinite set of numbers $\{a_n\}$.

A **series** is the sum of some, or all, members of a **sequence**, i.e. they can be **finite** or **infinite**.

e.g. $\sum_{n=1}^5 a_n$ is finite

$\sum_{n=1}^{\infty} b_n$ is infinite

Any **finite series** has a **finite value**, but **infinite series** can either be **convergent** to a finite value, or be **divergent**, i.e. tends to infinity.

For any **infinite series** $\sum_{n=0}^{\infty} a_n$, we can define:

The partial sum $S_n = \sum_{r=0}^n a_r = a_0 + a_1 + \dots + a_n$

The remainder $R_n = \sum_{r=n+1}^{\infty} a_r = a_{n+1} + a_{n+2} + \dots$

Such a series is **convergent** to a finite value S if:

$$\lim_{n \rightarrow \infty} S_n = S \Leftrightarrow \lim_{n \rightarrow \infty} R_n = 0$$

Note: Sum of arithmetic series $\sum_{k=0}^{n-1} (a + kd) = \frac{n}{2} (2a + (n-1)d)$

Sum of geometric series $\sum_{k=1}^n a_k r^{k-1} = \frac{a_1 (1-r^n)}{1-r}$

Sum of infinite geometric series $\sum_{k=1}^{\infty} a_k r^{k-1} = \frac{a_1}{1-r}$, $|a| < 1$

Sum of first n natural numbers $\sum_{k=1}^n k = \frac{1}{2} n(n+1)$

Example

Does $\sum_{n=0}^{\infty} \frac{1}{2^n}$ converge?

$\sum_{n=0}^{\infty} \frac{1}{2^n}$ is a geometric series with $a=1$, $r=\frac{1}{2}$

$$\therefore S_n = \frac{1 - (\frac{1}{2})^n}{1 - \frac{1}{2}} = 2(1 - (\frac{1}{2})^n) = 2 - \frac{1}{2^{n-1}}$$

$$\lim_{n \rightarrow \infty} S_n = 2$$

$\therefore \sum_{n=0}^{\infty} \frac{1}{2^n}$ converges to 2

If two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge to α and β respectively, then:

$$\sum_{n=1}^{\infty} (a_n + b_n) = \alpha + \beta$$

$$\sum_{n=1}^{\infty} (a_n - b_n) = \alpha - \beta$$

If a series $S = \sum_{r=1}^{\infty} a_r$ absolutely converges, i.e. if $T = \sum_{r=1}^{\infty} |a_r|$ converges, then S also converges.

Proof:

$$\text{Let } S_n = \sum_{r=1}^n a_r, \quad T_n = \sum_{r=1}^n |a_r|$$

For each r from 1 to n , we have:

$$a_r + |a_r| = 0 \text{ for } a_r < 0 \text{ (i.e. } a_r = -|a_r|)$$

$$a_r + |a_r| = 2|a_r| \text{ for } a_r \geq 0 \text{ (i.e. } a_r = |a_r|)$$

As a result, the partial sum $(S+T)_n$ has a value between 0 and $2 \times T_n$, i.e.:

$$0 \leq (S+T)_n \leq 2T_n$$

If T_n converges, then $\lim_{n \rightarrow \infty} T_n$ is finite. Let $\lim_{n \rightarrow \infty} T_n = t$:

$$0 \leq \lim_{n \rightarrow \infty} (S+T)_n \leq \underbrace{2t}_{\text{finite}}$$

Then, $\lim_{n \rightarrow \infty} (S+T)_n$ must also be finite as it is less than the finite value $2t$. Since $S_{\infty} = \sum_{r=1}^{\infty} a_r$, $T_{\infty} = \sum_{r=1}^{\infty} |a_r|$, this means

that $\sum_{r=1}^{\infty} (a_r + |a_r|)$ is convergent (if T is convergent). Therefore:

$$S = \sum_{r=1}^{\infty} a_r = \underbrace{\sum_{r=1}^{\infty} (a_r + |a_r|)}_{\text{Convergent}} - \underbrace{\sum_{r=1}^{\infty} |a_r|}_{\text{Convergent}}$$

$\therefore S$ converges if T converges.

Tests for Convergence/Divergence of Series

Simple test:

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=0}^{\infty} a_n$ is divergent.

Note, if $\lim_{n \rightarrow \infty} a_n = 0$, $\sum_{n=0}^{\infty} a_n$ could be either convergent or divergent; the test is inconclusive.

E.g. $\sum_{n=0}^{\infty} n-1$ is divergent since $\lim_{n \rightarrow \infty} (n-1) \neq 0$

Comparison Test

Let $\sum_{n=1}^{\infty} a_n$ be the series we want to test for convergence/divergence

Convergence:

Let $\sum_{n=1}^{\infty} c_n$ be a convergent series with all $c_1, c_2, \dots > 0$.

The series $\sum_{n=1}^{\infty} a_n$ absolutely converges if there exists an integer N such that:

$$|a_n| \leq c_n \text{ for all } n > N$$

If our series has a lesser value than one $\rightarrow$ known to be convergent, then it must also be convergent

Divergence:

Let $\sum_{n=1}^{\infty} d_n$ be a divergent series with all $d_1, d_2, \dots > 0$.

The series $\sum_{n=1}^{\infty} a_n$ diverges if there exists an integer N such that:

$$0 \leq d_n \leq a_n \text{ for all } n > N$$

If our series has a greater value than one $\rightarrow$ known to be divergent (i.e. go to infinity), then our series must also be divergent.

Examples

Does $\sum_{k=1}^{\infty} \frac{k}{k^3+1}$ Converge or diverge?

$a_k = \frac{k}{k^3+1} < \frac{1}{k^2}$ for all $k \geq 1$, and $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is known to converge

$\therefore \sum_{k=1}^{\infty} \frac{k}{k^3+1}$ Converges by the Comparison Test.

Note it is well-known that $\sum_{n=1}^{\infty} \frac{1}{k^n}$ diverges

and $\sum_{n=1}^{\infty} \frac{1}{k^{n^2}}$ Converges

Does $\sum_{k=1}^{\infty} \frac{1}{1+2 \ln k}$ Converge or diverge?

$a_k = \frac{1}{1+2 \ln k} > \frac{1}{1+2k} > \frac{1}{3k}$, and $\sum_{k=1}^{\infty} \frac{1}{3k} = \frac{1}{3} \sum_{k=1}^{\infty} \frac{1}{k}$ is known to diverge

$\therefore \sum_{k=1}^{\infty} \frac{1}{1+2 \ln k}$ diverges by the Comparison Test.

Ratio Test

Let $\sum_{n=1}^{\infty} a_n$ be the series we want to test for convergence/divergence, with all $a_n \neq 0$.

Let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$, then:

- If $L < 1$, $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

- If $L > 1$, $\sum_{n=1}^{\infty} a_n$ is divergent.

- If $L = 1$, the test is inconclusive.

Examples

Determine whether $\sum_{k=1}^{\infty} \frac{10^k}{k!}$ Converges or diverges

$$\begin{aligned} a_{n+1} &= \frac{10^{n+1}}{(n+1)!} \\ a_n &= \frac{10^n}{n!} \\ \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\frac{10^{n+1}}{(n+1)!}}{\frac{10^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{10^{n+1} n!}{10^n n! (n+1)} \right| = \lim_{n \rightarrow \infty} \left(\frac{10}{n+1} \right) \\ &= 0 < 1 \end{aligned}$$

$\therefore \sum_{k=1}^{\infty} \frac{10^k}{k!}$ Converges by the Ratio Test.

Determine whether $\sum_{n=1}^{\infty} \frac{69^n}{(-420)^{n+1} n}$ Converges or diverges

$$\begin{aligned} a_{n+1} &= \frac{69^{n+1}}{(-420)^{n+2} (n+1)} \\ a_n &= \frac{69^n}{(-420)^{n+1} n} \\ \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{69^{n+1} (-420)^{n+1} n}{69^n (-420)^{n+2} (n+1)} \right| = \lim_{n \rightarrow \infty} \left| \frac{69n}{420(n+1)} \right| = \frac{69}{420} \lim_{n \rightarrow \infty} \frac{n}{n+1} \\ &= \frac{69}{420} < 1 \end{aligned}$$

absolute value means the negative can disappear

$\therefore \sum_{n=1}^{\infty} \frac{69^n}{(-420)^{n+1} n}$ Converges by the Ratio Test.

n^{th} root test

Let $\sum_{n=1}^{\infty} a_n$ be the series we want to test for convergence/divergence.

Let $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$, then (as with the Ratio Test):

- If $L < 1$, $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

- If $L > 1$, $\sum_{n=1}^{\infty} a_n$ is divergent.

- If $L = 1$, the test is inconclusive.

Example

Determine whether $\sum_{k=0}^{\infty} \left(\frac{k}{2k+1}\right)^k$ converges or diverges

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

$$= \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{n}{2n+1}\right)^n} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \lim_{n \rightarrow \infty} \frac{1}{2 + 1/n} = \frac{1}{2} < 1$$

$\therefore \sum_{k=0}^{\infty} \left(\frac{k}{2k+1}\right)^k$ converges by the n^{th} Root Test.

Alternating Series Test

Let $\sum_{n=1}^{\infty} (-1)^n a_n$ be the series we want to test for convergence/divergence. (or $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$)

This series converges if:

- $a_n > 0$ (for all $n \in \mathbb{N}, n > 0$)

- $a_{n+1} < a_n$, i.e. the series is decreasing, for all $n > N$ for some finite $N \in \mathbb{N}$.

- $\lim_{n \rightarrow \infty} a_n = 0$

Examples

Determine whether $\sum_{k=1}^{\infty} \frac{(-1)^k}{69k}$ converges or diverges

note: a_k does not include the $(-1)^k$ (or $(-1)^{k+1}$) here.

$$a_k = \frac{1}{69k} > 0$$

$$a_{k+1} = \frac{1}{69(k+1)} = \frac{1}{69k+69} < a_k$$

$$\lim_{n \rightarrow \infty} \frac{1}{69n} = 0$$

$\therefore \sum_{k=1}^{\infty} \frac{(-1)^k}{69k}$ converges by Alternating Series Test

Determine if $\sum_{k=1}^{\infty} \frac{k \cos(\pi k)}{k^2+1}$ is convergent or divergent

$$\sum_{k=1}^{\infty} \cos(\pi k) = -1 + 1 - 1 + 1 - 1 + \dots = (-1)^k$$

$$\therefore \sum_{k=1}^{\infty} \frac{k \cos(\pi k)}{k^2+1} = \sum_{k=1}^{\infty} (-1)^k \frac{k}{k^2+1}$$

$$a_n = \frac{n}{n^2+1} > 0$$

$$a_{n+1} = \frac{n+1}{(n+1)^2+1} = \frac{n+1}{n^2+2n+2} = \frac{n(n+1)}{n^3+2n^2+2n}$$

$$a_n = \frac{n}{n^2+1} = \frac{n(n+1)}{n^3+n^2+n+1} = \frac{n(n+1)}{n^3+2n^2+1} > a_{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{n}{n^2+1} = \lim_{n \rightarrow \infty} \frac{1}{n + 1/n} = 0$$

$\therefore \sum_{k=1}^{\infty} \frac{k \cos(\pi k)}{k^2+1}$ is convergent by Alternating Series Test.

Power Series

A power series is a series involving a variable, x , and a constant, x_0 , of the form:

$$\sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 + a_1 (x-x_0) + a_2 (x-x_0)^2 + \dots$$

The power series is said to be centred around the point x_0 .

At a specific value of x , $(x-x_0)$ is just a constant, so the power series becomes a regular series. This means we can use our tests for convergence to see for which values of x is the series convergent:

Let $S = \sum_{n=0}^{\infty} a_n (x-x_0)^n$ be an arbitrary power series

From the Ratio Test:

$$\text{If } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (x-x_0)^{n+1}}{a_n (x-x_0)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (x-x_0)}{a_n} \right| < 1, \text{ we have convergence.}$$

Since $|x-x_0|$ is a constant (independent of n) we can take it out of the limit, i.e. we have convergence if:

$$|x-x_0| \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \rightarrow |x-x_0| < \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

As a result,

We define the radius of convergence, r , of a power series, S , to be:

$$r = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

Where S is convergent for all x such that $x_0 - r < x < x_0 + r$ and divergent for all x such that $x < x_0 - r$ or $x > x_0 + r$. At $x = x_0 \pm r$, we need to test separately, using any of our tests for convergence of regular series (except Ratio Test)

Example

Find the range of values of x for which $S = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x+1)^n}{n}$ converges

Get radius of convergence:

$$a_n = \frac{(-1)^{n+1}}{n}, \quad a_{n+1} = \frac{(-1)^{n+2}}{n+1}$$

$$r = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| -1 \times \frac{n+1}{n} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{n} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{1} = 1$$

$\therefore S$ converges for $-2 < x < 0$

Consider endpoints:

$$\text{For } x = -2, S = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(-1)^n}{n}$$

$$= - \sum_{n=1}^{\infty} \frac{1}{n} \text{ which diverges}$$

$$\text{for } x = 0, S = \sum_{n=1}^{\infty} (-1)^{n+1} \times \frac{1}{n}$$

Alternating Series test:

$$a_n = \frac{1}{n} \neq 0 \quad \checkmark$$

$$a_{n+1} = \frac{1}{n+1} < a_n \quad \checkmark$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \checkmark$$

$\therefore$ Convergent at $x = 0$ by Alternating Series Test.

$\therefore S$ converges for all x such that $-2 < x \leq 0$

Within the radius of convergence, $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is continuous and:

$$\left. \begin{aligned} - f'(x) &= \sum_{n=0}^{\infty} n a_n x^{n-1} \rightarrow \text{differentiate each term wrt } x \\ - \int f(x) dx &= \sum_{n=0}^{\infty} \frac{a_n x^{n+1}}{n+1} \rightarrow \text{integrate each term wrt } x \end{aligned} \right\} \text{ have the same radius of convergence as } f(x)$$

Taylor Series

Suppose a function, $f(x)$, is infinitely differentiable, and has a power series representation, i.e.:

$$f(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n = c_0 + c_1 (x-x_0) + c_2 (x-x_0)^2 + c_3 (x-x_0)^3 + \dots$$

To find this representation, we need to find $c_0, c_1, \dots$.

We can find c_0 by setting $x = x_0$:

$$c_0 = f(x_0)$$

Now, find $f'(x)$ by differentiating the power series termwise:

$$f'(x) = c_1 + 2c_2(x-x_0) + 3c_3(x-x_0)^2 + 4c_4(x-x_0)^3 + \dots$$

And we get:

$$c_1 = f'(x_0)$$

Applying this repeatedly gives:

$$f''(x) = 2c_2 + 3 \times 2 c_3 (x-x_0) + 4 \times 3 c_4 (x-x_0)^2 + \dots \Rightarrow c_2 = \frac{f''(x_0)}{2!}$$

$$f'''(x) = 3! c_3 + 4 \times 3 \times 2 c_4 (x-x_0) + \dots \Rightarrow c_3 = \frac{f'''(x_0)}{3!}$$

$\vdots$

This results in:

The Taylor Series representation of f is given by:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$
$$= f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!} (x-x_0)^2 + \dots$$

With the n th degree Taylor polynomial being the partial sum of this series, i.e.:

$$f(x) \approx \sum_{i=0}^n \frac{f^{(i)}(x_0)}{i!} (x-x_0)^i$$
$$= f(x_0) + f'(x_0)(x-x_0) + \dots + \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$

Note: The special case when $x_0 = 0$ is called the Maclaurin Series.

Example

Find the Taylor series for $f(x) = \ln(3+4x)$ about $x = 1/4$

$$f(x) = \ln(3+4x)$$

$$f'(x) = 4(3+4x)^{-1}$$

$$f''(x) = -4^2(3+4x)^{-2}$$

$$f'''(x) = 4^3(2)(3+4x)^{-3}$$

$$f^{(4)}(x) = -4^4(2 \times 3)(3+4x)^{-4}$$

$$f^{(5)}(x) = 4^5(2 \times 3 \times 4)(3+4x)^{-5}$$

$$\text{In general, } f^{(n)}(x) = (-1)^{n+1} 4^n (n-1)! (3+4x)^{-n}, \quad n > 0$$

At $x = 1/4$:

$$f(1/4) = \ln 4$$

$$f^{(n)}(1/4) = (-1)^{n+1} (n-1)! , \quad n > 0$$

Taylor Series formula: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$

$$\ln(3+4x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(1/4)}{n!} (x-1/4)^n$$

$$= \ln 4 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n-1)!}{n!} (x-1/4)^n$$

$$= \ln 4 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1/4)^n$$

$\therefore$ Taylor series of $f(x)$ centred about $x=1/4$ is $\ln 4 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1/4)^n$

Fourier Series

For a function $f(x)$, to have a Taylor Series representation around x_0 , f must be infinitely many times differentiable at x_0 . If this is not the case, we can instead use Fourier Series to represent f using trigonometric functions.

Suppose that a series exists such that for a function $f(x)$, $-\pi < x < \pi$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

Integrating over $(-\pi, \pi)$ gives:

$$\int_{-\pi}^{\pi} f(x) dx = \frac{a_0}{2} \int_{-\pi}^{\pi} 1 dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos(nx) dx + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin(nx) dx$$

$\parallel \quad \parallel$
 $0 \quad 0$

$$\pi a_0 = \int_{-\pi}^{\pi} f(x) dx$$

Trig graphs between $-\pi$ and π have the same area beneath the x axis as above it, i.e. $\int_{-\pi}^0 \cos x dx = -\int_0^{\pi} \cos x dx \Rightarrow \int_{-\pi}^{\pi} \cos x dx = 0$ and same for $\sin x$. The same is not true for $\cos^2 x$ and $\sin^2 x$

Considering $\int_{-\pi}^{\pi} f(x) \cos(mx) dx$, $m > 0$, gives:

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \cos(mx) dx &= \int_{-\pi}^{\pi} \left(\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx) \right) \cos(mx) dx \\ &= \frac{a_0}{2} \int_{-\pi}^{\pi} \cos(mx) dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx \end{aligned}$$

$\parallel \quad \downarrow \quad \parallel$
 $0 \quad \begin{cases} 0 & \text{if } n \neq m \\ \pi & \text{if } n = m \end{cases} \quad 0$

$$a_m \int_{-\pi}^{\pi} \cos^2(mx) dx = \int_{-\pi}^{\pi} f(x) \cos(mx) dx$$

$$\pi a_m = \int_{-\pi}^{\pi} f(x) \cos(mx) dx \quad \text{for all } m > 0$$

Considering $\int_{-\pi}^{\pi} f(x) \sin(mx) dx$, $m > 0$, gives:

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \sin(mx) dx &= \int_{-\pi}^{\pi} \left(\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx) \right) \sin(mx) dx \\ &= \frac{a_0}{2} \int_{-\pi}^{\pi} \sin(mx) dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos(nx) \sin(mx) dx + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx \end{aligned}$$

$\parallel \quad \parallel \quad \downarrow$
 $0 \quad 0 \quad \begin{cases} 0 & \text{if } n \neq m \\ \pi & \text{if } n = m \end{cases}$

$$b_m \int_{-\pi}^{\pi} \sin^2(mx) dx = \int_{-\pi}^{\pi} f(x) \sin(mx) dx$$

$$\pi b_m = \int_{-\pi}^{\pi} f(x) \sin(mx) dx$$

The result of these gives the definition:

For a function $f(x)$, $-\pi < x < \pi$, the Fourier Series is:

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

Where:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

At all points x at which f is continuous, the series converges to $f(x)$

At any points y where there is a jump in discontinuity, the series converges to $\frac{1}{2} \left(\lim_{x \rightarrow y^-} f(x) + \lim_{x \rightarrow y^+} f(x) \right)$, i.e. the average of the jump.

Note: if we have a function $g(x)$, $a < x < b$ we can define:

$$f(x) = g\left(x - \frac{b-a}{2} \frac{2\pi}{b-a}\right)$$

Where $f(x)$ has range $-\pi < x < \pi$ and so can be used to find the Fourier Series expansion for g .

Example

Find the Fourier Series expansion for $y = x^2 + x$ over $(-\pi, \pi)$

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

$$\begin{aligned} \frac{a_0}{2} &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 + x) dx \\ &= \frac{1}{\pi} \left[\frac{1}{3} x^3 + \frac{1}{2} x^2 \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left(\frac{1}{3} \pi^3 + \frac{1}{2} \pi^2 - \left(\frac{1}{3} \pi^3 - \frac{1}{2} \pi^2 \right) \right) = \frac{2\pi^2}{3} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 \cos(nx) + x \cos(nx)) dx \end{aligned}$$

$$x \cos(nx) \text{ is an odd function so } \int_{-\pi}^{\pi} x \cos(nx) dx = 0$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(nx) dx$$

$$\text{Integration by parts } \rightarrow u = x^2, v' = \cos(nx) \\ u' = 2x, v = \frac{1}{n} \sin(nx)$$

$$= \frac{1}{\pi} \left[\frac{x^2}{n} \sin(nx) - \frac{2}{n} \int x \sin(nx) dx \right]_{-\pi}^{\pi}$$

$$\text{Integration by parts } \rightarrow u = x, v' = \sin(nx) \\ u' = 1, v = -\frac{1}{n} \cos(nx)$$

$$= \frac{1}{\pi} \left[\frac{x^2}{n} \sin(nx) - \frac{2}{n} \left(-\frac{x}{n} \cos(nx) + \frac{1}{n} \int \cos(nx) dx \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{x^2}{n} \sin(nx) + \frac{2x}{n^2} \cos(nx) - \frac{2}{n^3} \sin(nx) \right]_{-\pi}^{\pi}$$

a function $f(x)$ is odd if $f(-x) = -f(x)$

In this case $\int_{-a}^a f(x) dx = 0$

$$\text{Using } \sin(n\pi) = 0, \cos(n\pi) = (-1)^n :$$

$$= \frac{1}{\pi} \left[0 + \frac{2\pi}{n^2} (-1)^n - 0 - 0 + \frac{2\pi}{n^2} (-1)^n + 0 \right]$$

$$a_n = \frac{4}{n^2} (-1)^n$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin(nx) + x \sin(nx) dx$$

$$\text{Since } x^2 \sin(nx) \text{ is odd, } \int_{-\pi}^{\pi} x^2 \sin(nx) dx = 0$$

$$\therefore b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$$

$$= \frac{1}{\pi} \left[-\frac{x}{n} \cos(nx) - \frac{1}{n^2} \sin(nx) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left(-\frac{\pi}{n} (-1)^n - \frac{\pi}{n} (-1)^n \right)$$

$$= -\frac{2}{n} (-1)^n = \frac{2}{n} (-1)^{n+1}$$

$$\therefore f(x) \approx \frac{2\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos(nx) + \frac{2}{n} (-1)^{n+1} \sin(nx)$$